FLAT PULLBACK PRESERVES RATIONAL EQUIVALENCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xypic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Local finiteness of the inverse image of a locally finite family. Same chapter chow.tex, label lemma-inverse-image-locally-finite, invoked at proof line 3195. Makes the pulled back family of integral subschemes locally finite in the source, so that the sum makes sense.
- Flat pullback versus proper pushforward. Same chapter chow.tex, label lemma-flat-pullback-proper-pushforward, invoked at proof line 3203. Rewrites the pullback of the sum of pushforwards as the sum of pushforwards of the pulled back divisors over the fibre products.
- Pullback of a divisor of a rational function along a flat morphism, and infinite sums of rational equivalences. Same chapter chow.tex, labels lemma-prepare-flat-pullback-rational-equivalence (proof line 3208) and remark-infinite-sums-rational-equivalences (proof line 3213). The first shows each pulled back principal divisor is rationally trivial on its fibre product, and the second lets the local finiteness upgrade that to the whole sum.

Source reference. Original lemma 02RG: sources/stacks-chow/pinned-original-source.tex, source lines 2401–2401. Navigation pointer for \ref{lemma-flat-pullback-proper-pushforward} used inside the counted span (line 3203); the argument that consumes it is retained in the span.

Source reference. Original lemma 02R9: sources/stacks-chow/pinned-original-source.tex, source lines 2170–2170. Navigation pointer for \ref{lemma-inverse-image-locally-finite} used inside the counted span (line 3195); the argument that consumes it is retained in the span.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0EPH: sources/stacks-chow/pinned-original-source.tex, source lines 3098–3098. Navigation pointer for \ref{lemma-prepare-flat-pullback-rational-equivalence} used inside the counted span (line 3208); the argument that consumes it is retained in the span.

Source reference. Original remark 02RZ: sources/stacks-chow/pinned-original-source.tex, source lines 3057–3057. Navigation pointer for \ref{remark-infinite-sums-rational-equivalences} used inside the counted span (line 3213); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 3174); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X, Y be schemes locally of finite type over S. Let $f : X \to Y$ be a flat morphism of relative dimension r. Let $\alpha \sim_{rat} \beta$ be rationally equivalent k-cycles on Y. Then $f^*\alpha \sim_{rat} f^*\beta$ as $(k + r)$-cycles on X.*

Proof. What do we have to show? Well, suppose we are given a collection

$$i_j : W_j \longrightarrow Y$$

of closed immersions, with each W_j integral of δ-dimension $k + 1$ and rational functions $g_j \in R(W_j)^*$. Moreover, assume that the collection $\{i_j(W_j)\}_{j \in J}$ is locally finite on Y. Then we have to show that

$$f^*(\sum i_{j,*}\mathrm{div}(g_j)) = \sum f^* i_{j,*}\mathrm{div}(g_j)$$

is rationally equivalent to zero on X. The sum on the right makes sense as $\{W_j\}$ is locally finite in X by Lemma 02R9.

Consider the fibre products

$$i'_j : W'_j = W_j \times_Y X \longrightarrow X.$$

and denote $f_j : W'_j \to W_j$ the first projection. By Lemma 02RG we can write the sum above as

$$\sum i'_{j,*}(f_j^*\mathrm{div}(g_j))$$

By Lemma 0EPH we see that each $f_j^*\mathrm{div}(g_j)$ is rationally equivalent to zero on W'_j. Hence each $i'_{j,*}(f_j^*\mathrm{div}(g_j))$ is rationally equivalent to zero. Then the same is true for the displayed sum by the discussion in Remark 02RZ. □